Lab Project

EE 457 Fall 2024

Lab #3

Justin Pulley

For: Prof. Thuc

Date: 11/15/24

Purpose of Description

The purpose of this lab is to understand the theoretical foundations of Analog Communications as well as of Double-Sideband Amplitude Modulation (DSB AM) and Demodulation. To design the Simulink models of the DSB AM to analyze each signal in time and frequency domains using time scope and spectrum analyzer. To examine the effects of the Additive White Gaussian Noise (AWGN) in the Simulink model of DSB AM. To observe the real-time music transmission for DSB AM modulated signal via Universal Software Radio Peripheral (USRP) trans-receiver.

Design and Implementation Procedure

First, I used MATLAB to demonstrate Amplitude Modulation (AM) with $m(t) = \sin(16\pi t)$, $A_c = 3$, $\mu = 0.9$, $f_c = 60$ Hz, with the time domain of the signal is from -T/2 to T/2-dt where T = 1 and dt = 1e-3.

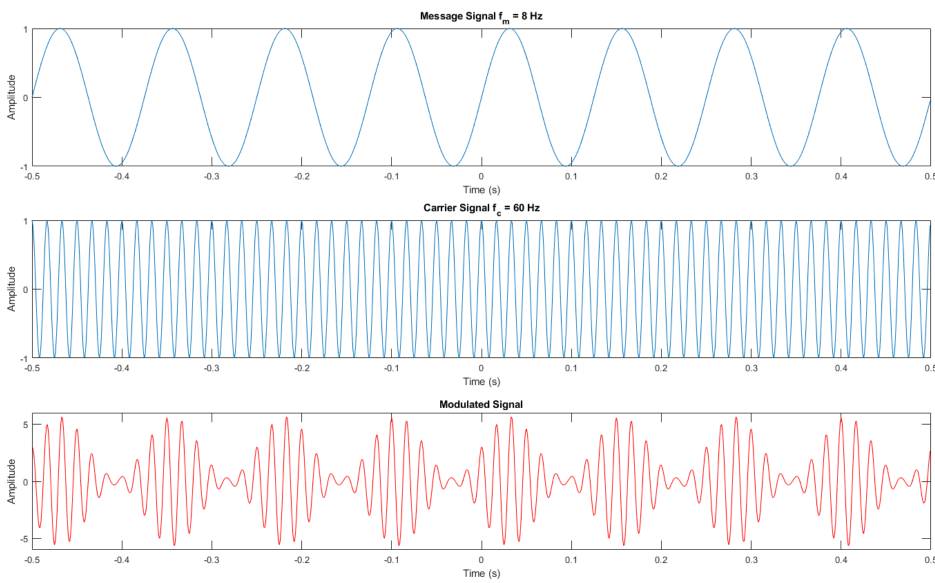

Figure 1: A plot of the message signal, carrier signal, and amplitude modulated signal all in the time domain.

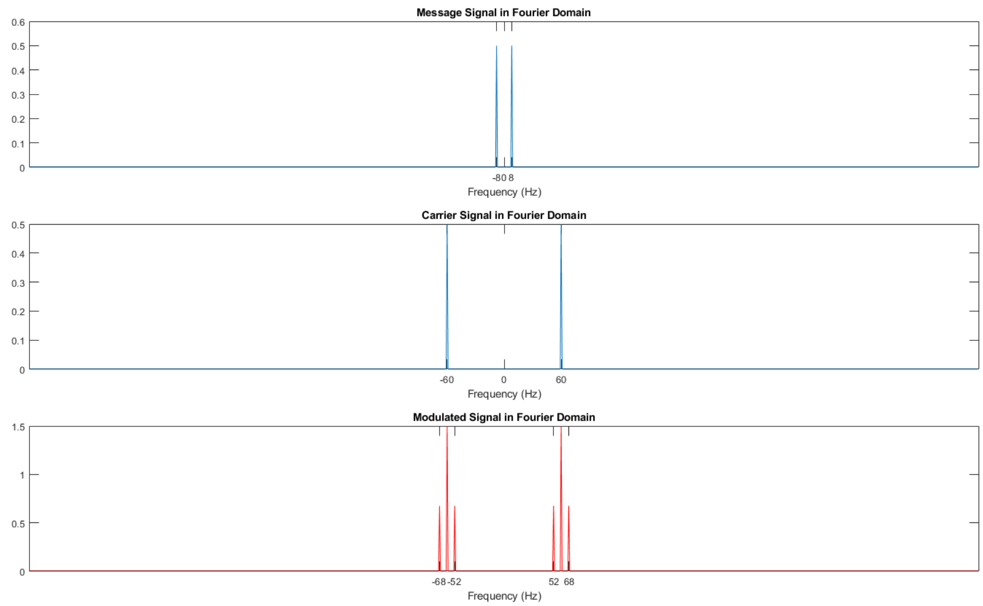

Figure 2: A plot of the message signal, carrier signal, and amplitude modulated signal all in the frequency domain.

Next, I built a Simulink model of Double Sideband Amplitude Modulation.

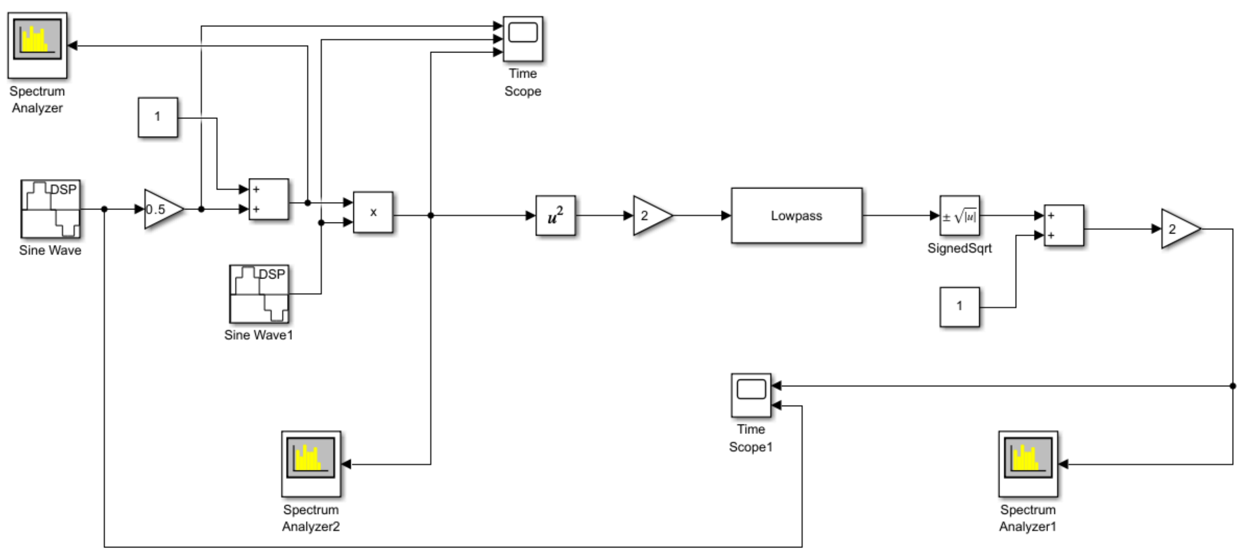

Figure 3: The Simulink model of Double Sideband Amplitude Modulation.

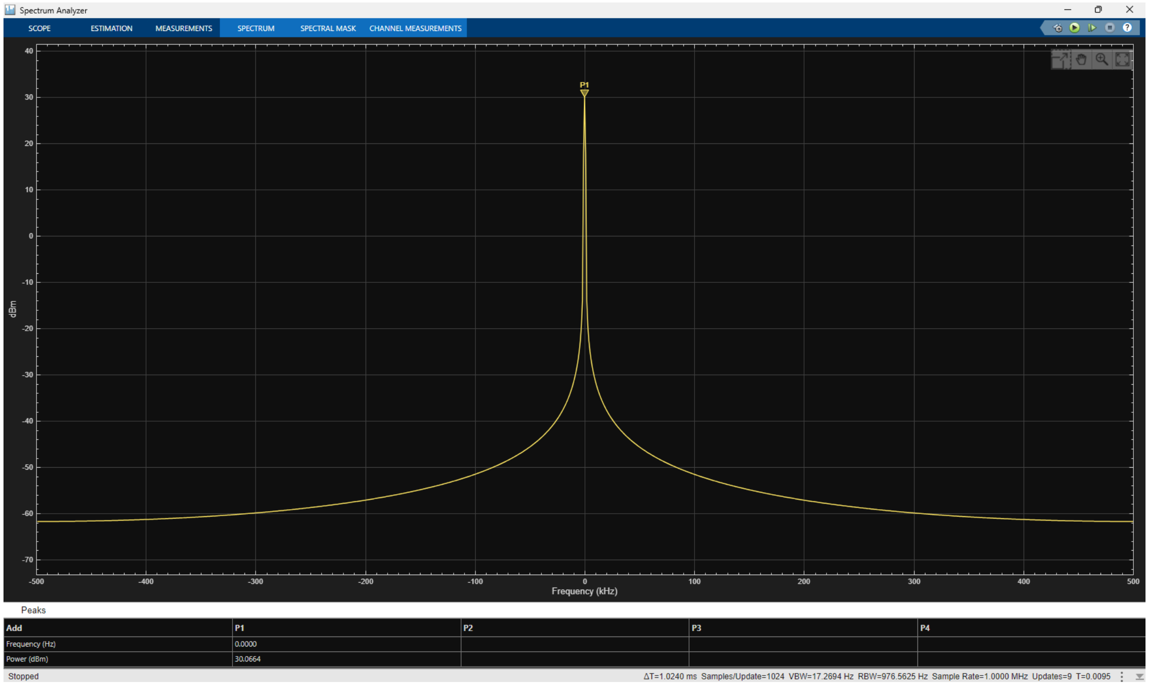

Figure 4: Plot of Spectrum Analyzer from Simulink model of Double Sideband Amplitude Modulation

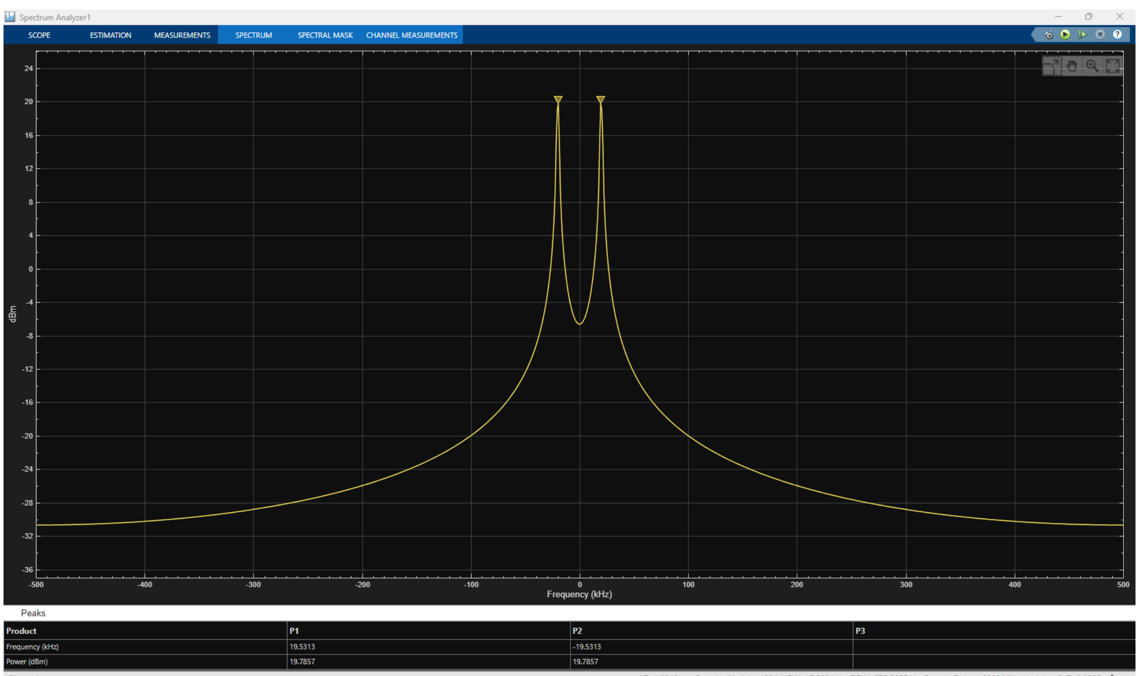

Figure 5: Plot of Spectrum Analyzer 1 from Simulink model of Double Sideband Amplitude Modulation

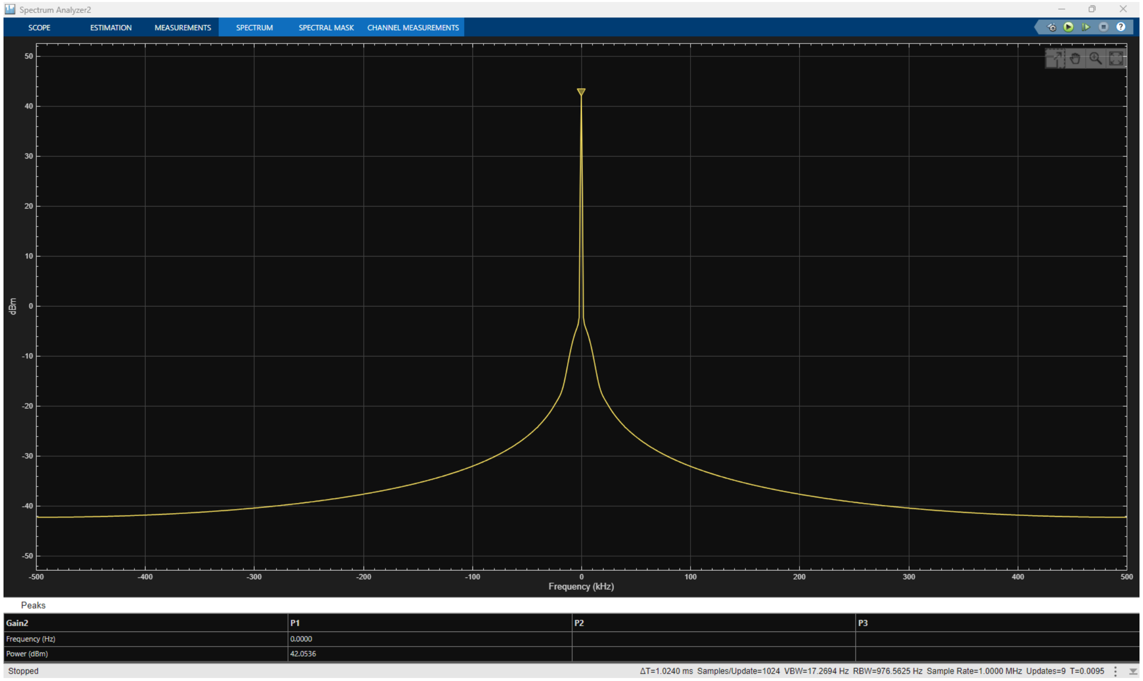

Figure 6: Plot of Spectrum Analyzer 2 from Simulink model of Double Sideband Amplitude Modulation

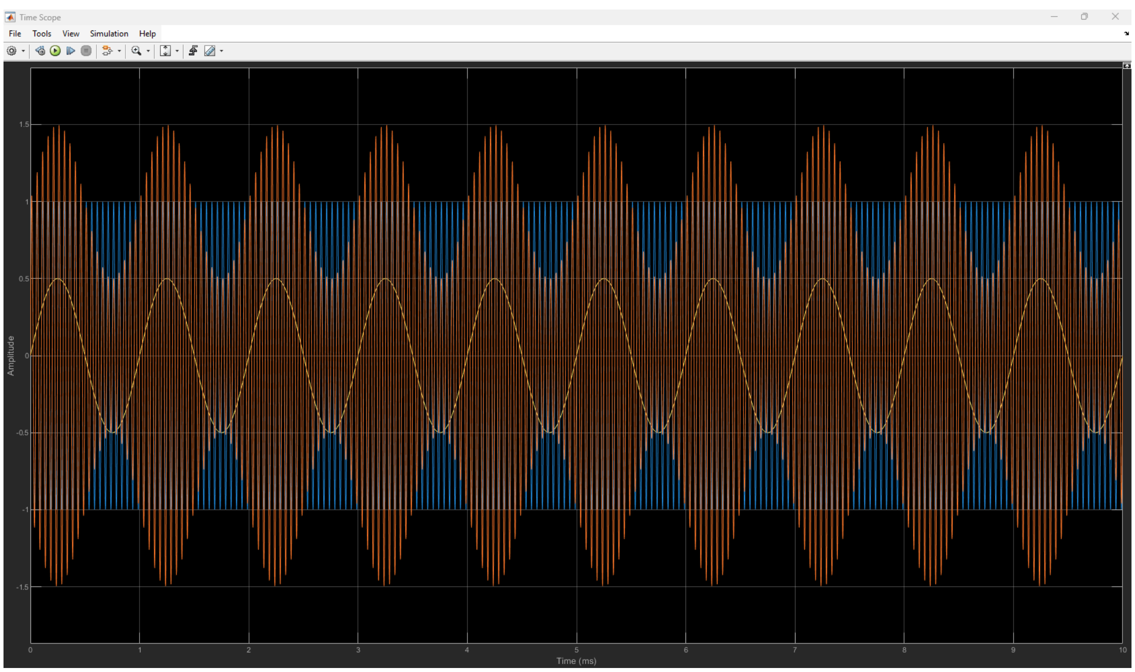

Figure 7: Plot of Time Scope from Simulink model of Double Sideband Amplitude Modulation

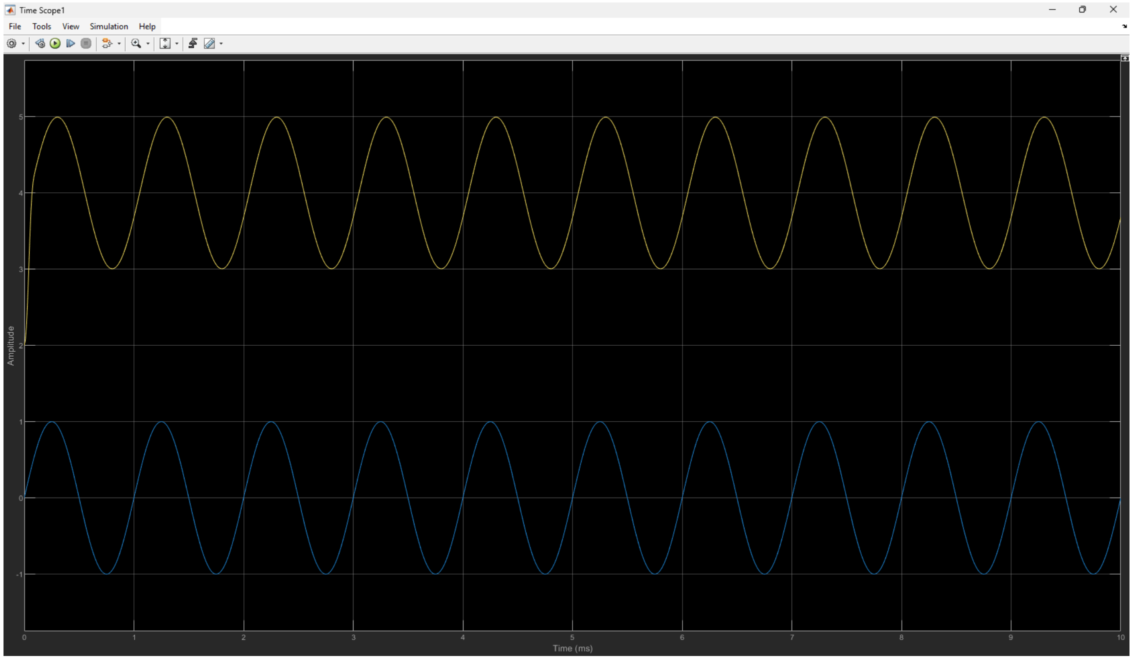

Figure 8: Plot of Time Scope 1 from Simulink model of Double Sideband Amplitude Modulation

Next, I built a Simulink model of music transmission using a DSB AM modulator and demodulator.

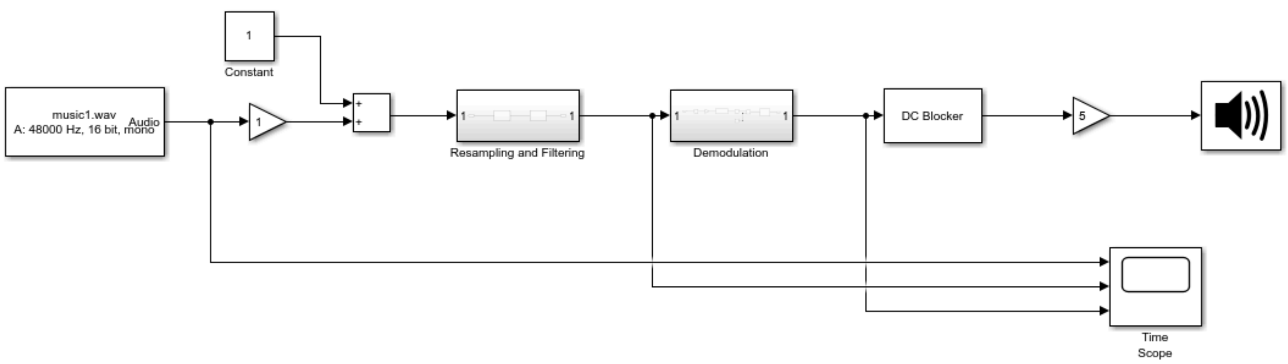

Figure 9: The Simulink model of music transmission using a DSB AM modulator and demodulator

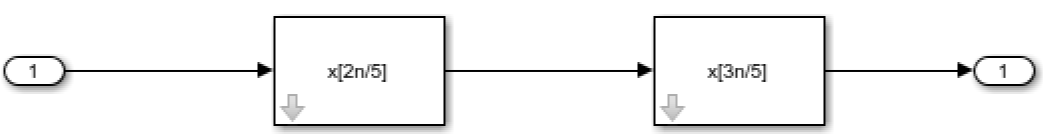

Figure 9.1: Resampling and Filtering Subsystem model.

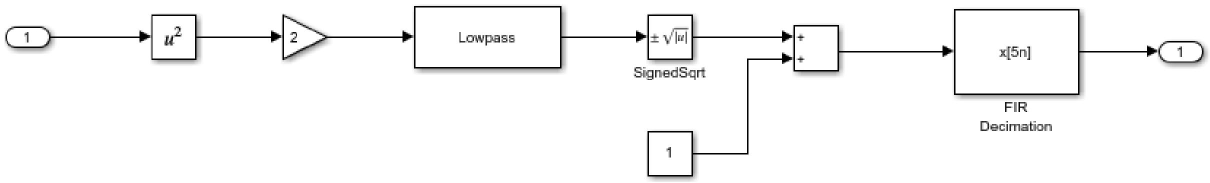

Figure 9.2: Demodulation Subsystem model.

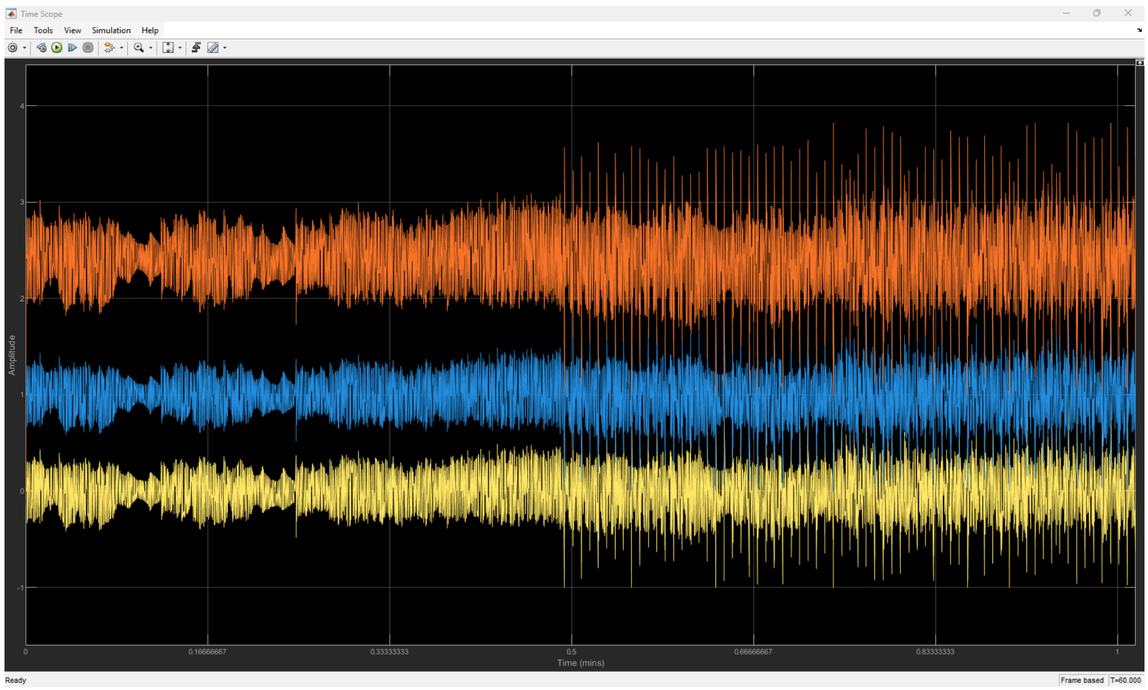

Figure 10: Plot of Time Scope from Simulink model of music transmission using DSB AM modulator and demodulator

Afterwards, I built a multimedia file transmitter and receiver using DSB AM via USRP hardware in Simulink. Using two computers connected to two USRP B200 boards, I ran a real-time transmission from one B200 board acting as a transmitter to the other board acting as a receiver. I successfully was able to complete the transmission of the multimedia file.

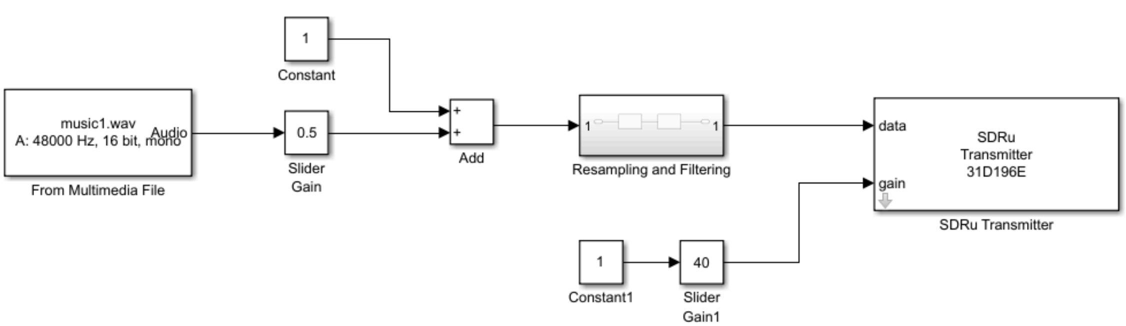

Figure 11: Simulink model of transmitter transmitting a Multimedia File Using DSB AM via USRP

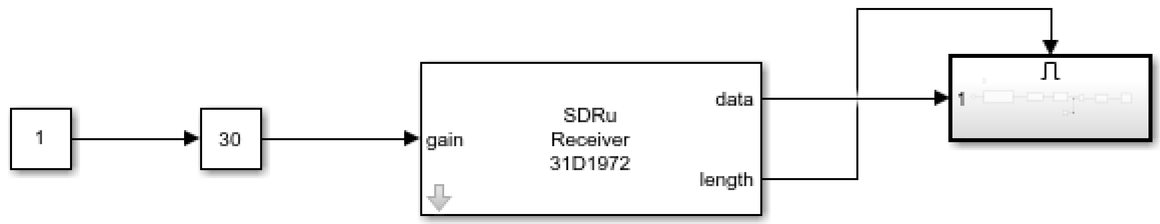

Figure 12: Simulink model of a receiver receiving a Multimedia File Using DSB AM via USRP

FInally, I built another Simulink model shown below.

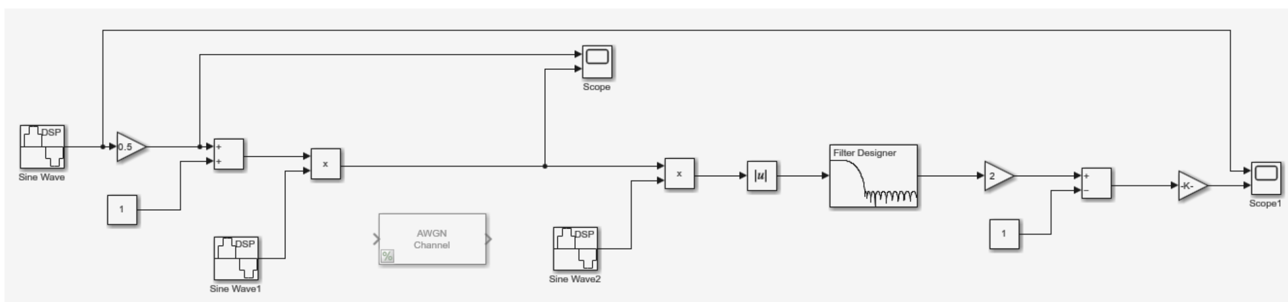

Figure 13: Simulink model of amplitude modulation and demodulation

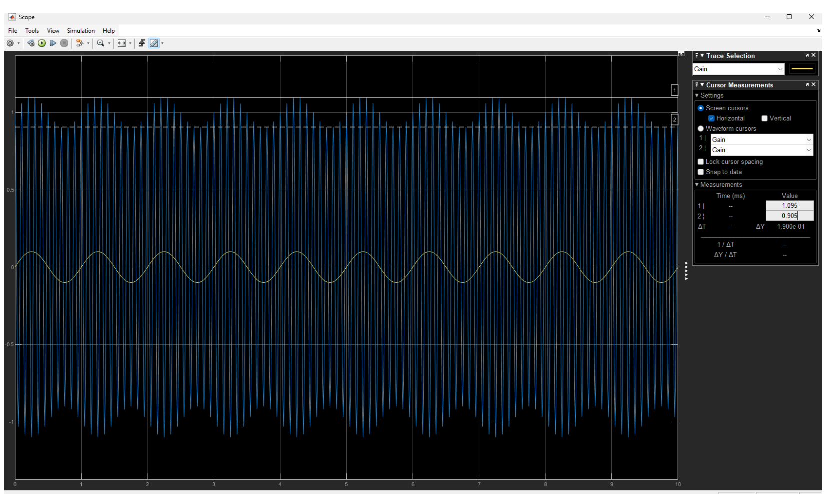

Figure 14: Output of Scope showing the message and amplitude modulated signal. The modulation index was set to 0.1.

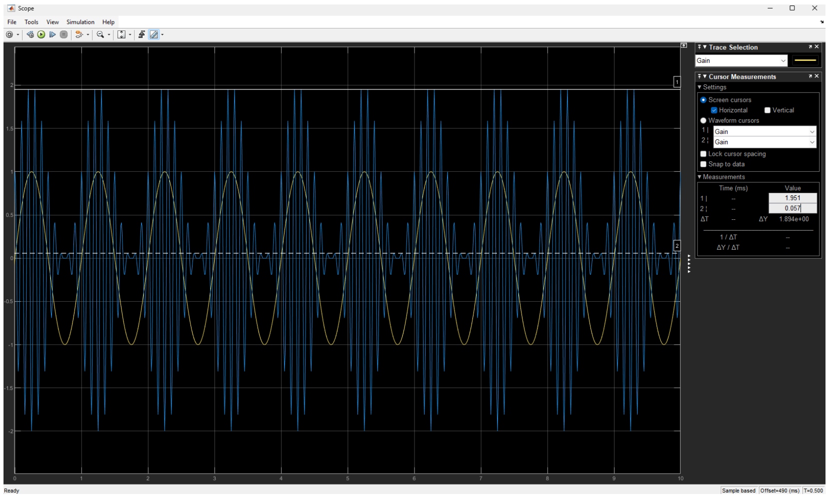

Figure 15: Output of Scope showing the message and amplitude modulated signal. The modulation index was set to 1.

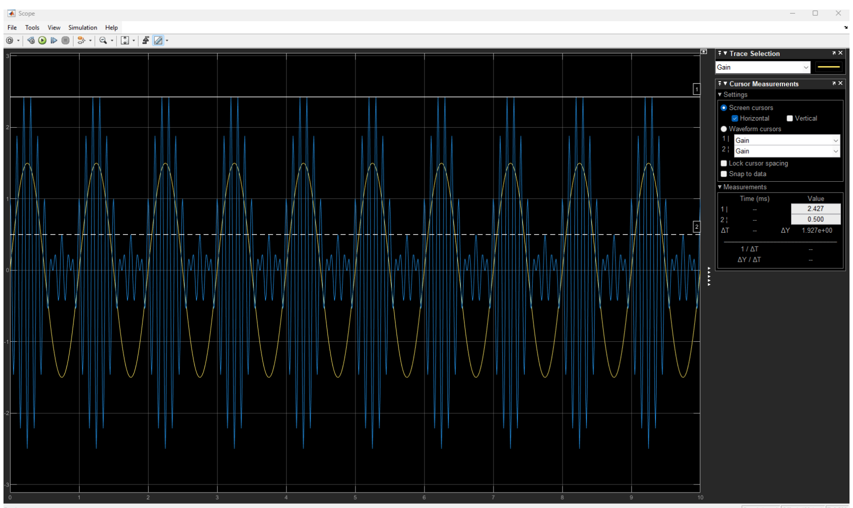

Figure 16: Output of Scope showing the message and amplitude modulated signal. The modulation index was set to 1.5. You can see the AM signal is distorted.

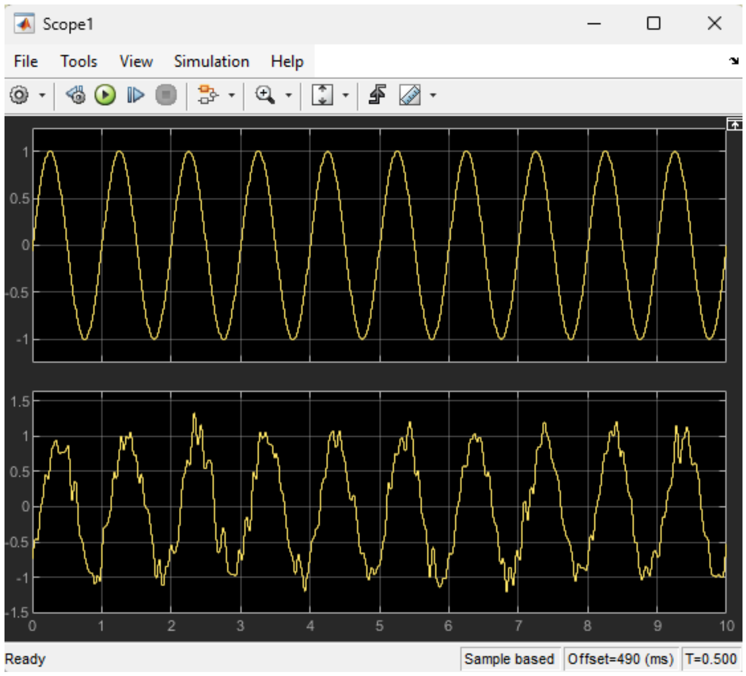

Figure 17: Output of Scope1 showing the message and demodulated signal. The noise variance was set to 0.01.

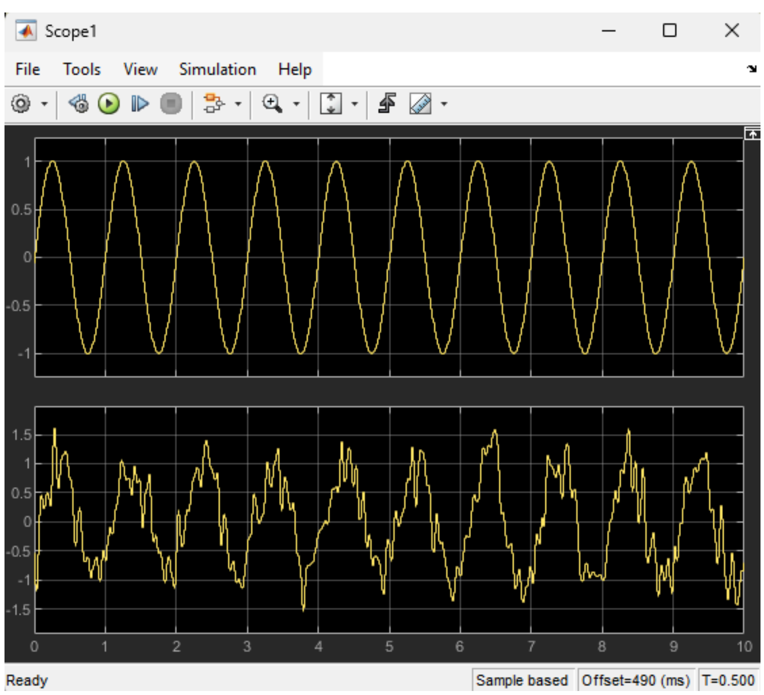

Figure 18: Output of Scope1 showing the message and demodulated signal. The noise variance was set to 0.05.

Observations

For Figure 13, the message signal and the demodulated signal have a slight phase difference due to the multiplication process inherent in amplitude modulation. During modulation, the message signal is multiplied by the carrier wave, leading to a small phase shift in the modulated signal.

Regarding Figure 13, I studied the effects of different modulation indexes. I ran the simulink model under three different modulation indexes: 0.1, 1, and 1.5. As I increased the modulation index, the minimum amplitude of the modulation signal would decrease. I found that when the modulation index is greater than 1, then the demodulated signal cannot be performed correctly. Also, I studied the effects of noise on the demodulated signal by setting the noise variance to 0.01 and 0.05. You can see that as you increase the variance the more distorted the demodulated signal is.

Lessons Learned

Throughout this lab, I learned the theoretical foundations of Analog Communications. I learned how to design the Simulink models of the DSB AM to analyze each signal in time and frequency domains using time scope and spectrum analyzer. I learned how to examine the effects of the Additive White Gaussian Noise (AWGN) in the Simulink model of DSB AM. I learned how to observe the real-time music transmission for DSB AM modulated signals via Universal Software Radio Peripheral (USRP) trans-receiver.